THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** interparticle forces of attraction	Weaker or stronger by state
2	**(b)** shape and volume	Strong forces, no free movement
3	**(d)** move past each other within a space	Slightly more space
4	**(c)** melting point	Solid to liquid
5	**(c)** 133 °C	Textbook table
6	**(b)** mixture of gases	Nitrogen, oxygen, argon and others
7	**(c)** a metal with other metals or non-metals	Solid mixtures
8	**(b)** 4 : 1	Tamra 4 parts and Vanga 1 part
9	**(d)** cannot be broken down into simpler substances	Building blocks of matter
10	**(a)** compound	New substance formed

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Perfume particles mix with air particles (1). Gas particles move freely (1).	1+1
12	They vibrate more vigorously (1). Interparticle forces weaken and the solid melts (1).	1+1
13	Iron + Sulfur (1). → Iron sulfide (1).	1+1
14	Elements with properties between metals and non-metals (1). Silicon and boron (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Particles are free and move apart (1½). They spread to fill the whole volume (1½).	1½+1½
16	Magnesium + Oxygen → Magnesium oxide (1). Magnesium and oxygen are elements (1). Magnesium oxide is a compound (1).	1+1+1
17	Solid wax melts into liquid (1). Liquid wax rises through wick and vaporises (1). Particle forces weaken at each stage (1).	1+1+1
	OR: Iron strongest (1). Urea next (1). Ice weakest (1).	1+1+1
18	Gold is a metallic element (1½). It occurs naturally as a pure solid, so it is a native mineral (1½).	1½+1½
	OR: Water: drinking, cleaning (1½). Carbon dioxide: fire extinguishers, cold drinks (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ice particles are farther apart (2). Less dense than water (1). Floats (1). Exception to solids being denser (1).	2+1+1+1
	OR: Flour is a solid (1). Grains keep shape (1). Pours like a liquid because tiny grains slide (1). Does not have definite volume as a liquid (1). Check by particles (1).	1+1+1+1+1
20	Use magnet to remove iron (1). Add water, sulfur floats (1). Sand sinks (1). Filter and dry (1). Name the physical processes (1).	1+1+1+1+1
	OR: Sugar is a single pure substance (1). Burning causes a chemical change (1). Products are different substances (1). Shows it is a compound (1). Mixture components can be separated physically (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** evaporation	A surface process
2	**(a)** are free and widely spaced	Large space
3	**(c)** solid to liquid	At 0 °C
4	**(c)** liquid and gas	Particles move past each other
5	**(b)** move faster	Greater energy
6	**(d)** iron sulfide	A new compound
7	**(a)** metalloids	Intermediate properties
8	**(a)** a naturally occurring element that is a metal	Native mineral
9	**(d)** Water, carbon dioxide, sodium chloride	Chemically combined elements
10	**(d)** They have properties different from their elements	New substance

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iron has stronger interparticle forces (1). More energy is needed to loosen the particles (1).	1+1
12	Slow change of liquid to vapour (1). Occurs below boiling point from the surface (1).	1+1
13	Natural solid substances of fixed chemical composition (1). Mostly compounds, rarely elements (1).	1+1
14	Minerals that are pure elements (1). Example: gold (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Particles need enough energy to escape (1½). At room temperature only slow evaporation occurs (1½).	1½+1½
16	It has water, fat, protein, sugar (1). They are not chemically combined (1). Composition varies (1).	1+1+1
17	Space decreases (1). Particles move slowly and come closer (1). Forces become more effective (1).	1+1+1
	OR: Camphor particles move into air (1). Air particles move freely and mix (1). Particles reach all corners (1).	1+1+1
18	Sugar solution can be made sweet or weak (1½). Amount of sugar can change (1½).	1½+1½
	OR: Alloys have improved strength (1). Better resistance to corrosion (1). Pure metals often too soft (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Gas particles move randomly (1). They hit the walls (1). Heating makes them faster (1). More hits so higher pressure (1). Cooling reduces pressure (1).	1+1+1+1+1
	OR: Evaporation at surface at any temperature (1). Boiling throughout at boiling point (1). Both turn liquid to vapour (1). Boiling is fast, evaporation slow (1). Particles gain enough energy to escape (1).	1+1+1+1+1
20	Sodium is a reactive metal (1). Chlorine is a poisonous gas (1). Sodium chloride is edible salt (1). Hydrogen and oxygen are gases, water is liquid (1). New substances form (1).	1+1+1+1+1
	OR: Element: copper wire (1). Compound: sodium chloride (1). Mixture: air (1). Alloy: steel (1). Mineral resources (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** solids	Particles are closely packed
2	**(a)** definite volume but no fixed shape	Particles move within space
3	**(c)** its particles are already close with little space	Little interparticle space
4	**(a)** boiling point	Liquid to vapour
5	**(c)** iron	Highest melting point
6	**(b)** components are evenly distributed	Same composition throughout
7	**(a)** copper and zinc	Example of alloy
8	**(c)** mixture (alloy)	Developed alloys
9	**(c)** Oxygen	Cannot be broken down
10	**(d)** fixed ratio	Fixed composition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solid: close, regular (1). Liquid: close, irregular; gas: far apart (1).	1+1
12	Milk is liquid with no fixed shape (1). Glass is solid with fixed shape (1).	1+1
13	It contains several gases (1). They are not chemically combined and keep their properties (1).	1+1
14	Ancient mixture of two or more metals (1). Had properties distinct from the metals (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solids: iron, ice, wood (1). Liquids: water, milk, oil (1). Gases: air, oxygen, methane (1).	1+1+1
16	Most rocks are mixtures of minerals (1). Minerals have fixed composition (1). Some like gold are native elements (1).	1+1+1
17	Particles move randomly (1). They collide with walls (1). Collisions produce pressure (1).	1+1+1
	OR: Yes as ice, water, vapour (1). Same particles in different arrangements (1). Depends on temperature (1).	1+1+1
18	Hydrogen (1). Word equation: Iron + HCl → Iron chloride + Hydrogen (1). Burns with a pop sound with a flame (1).	1+1+1
	OR: Distilled water has only water particles (1½). Tap water has dissolved salts and gases (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solid wax in candle (1). Liquid wax in wick pool (1). Vapour burns near flame (1). Particle picture each state (2).	1+1+1+2
	OR: Gas particles have negligible attraction (1). Move freely (1). Fill all space (1). Liquid particles attract and stay together (1). Settle under gravity at the bottom (1).	1+1+1+1+1
20	Iron gives hydrogen (1). Iron sulfide gives hydrogen sulfide (1). Products differ (1). Shows iron sulfide is a new compound (1). Observation of gas and its smell (1).	1+1+1+1+1
	OR: Between metals and non-metals (1). Intermediate properties (1). Silicon (1). Boron (1). Uses in electronics or glass (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** iron nail	Solid state
2	**(b)** are tightly held in place	Strong attraction
3	**(a)** water vapour (gas)	At 100 °C at atmospheric pressure
4	**(d)** liquid	Takes shape of container
5	**(b)** 100 °C	Liquid to vapour
6	**(d)** new substance with different properties	Compound differs from components
7	**(a)** natural solid substances with fixed chemical composition	Found in Earth
8	**(c)** compound (iron oxide)	Iron reacts with moist air
9	**(b)** Distilled water	Single substance
10	**(d)** carbon dioxide and water	Both contain only non-metal elements

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Particles of ice are farther apart than in water (1). So ice is less dense and floats (1).	1+1
12	Salt particles are very tiny (1). They spread uniformly among water particles (1).	1+1
13	Hydrogen (1). Iron + dilute HCl → iron chloride + hydrogen (1).	1+1
14	Mixtures have variable composition; compounds fixed (1). Mixture components keep properties; compound has new properties (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Particles vibrate more (1½). They move slightly apart, increasing size (1½).	1½+1½
16	It would not extinguish fire since hydrogen burns (1½). It would not have properties such as being a liquid solvent at room temperature (1½).	1½+1½
17	Ink and water particles move (1). They mix with each other (1). Ink spreads uniformly with time (1).	1+1+1
	OR: Chair is made of matter, which is made of particles (1½). Nothing would remain (1½).	1½+1½
18	Sugar (1). Baking soda (1). Salt (1).	1+1+1
	OR: Metal: iron (1). Non-metal: sulfur (1). Metalloid: silicon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Dissolving salt, no salt visible (1½). Diluting ink with water, colour persists (1½). Smell spreads over a large area (2).	1½+1½+2
	OR: Particles (1). Forces (1). Shape and volume (1). Temperature effect (1). Change of state (1).	1+1+1+1+1
20	Air is a mixture (1). Gases: nitrogen, oxygen, argon, CO_2 and water vapour (2). Liquefaction can separate components (1). Components keep properties (1).	1+2+1+1
	OR: Alloys improve strength (1). Stainless steel (1). Minerals provide metals (1). Compounds such as oxides are processed (1). Pure metals extracted from minerals (1).	1+1+1+1+1

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